

Taking Transmission Technology to a new standard

If you thought cramming for
exams was tough, try
cramming 50 percent
more information down
a fibre-optic cable.

By Sharon Ng
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While not exactly household terms,
the H.264 Standard is all around
us and powering our electronic
communications world. Thanks to
City University of Hong Kong
(CityU) researchers, that
standard is about
to lift.

H.264 is one of several fast motion estimation algorithms of core video encoding compression technology widely used in video conferencing, video telephony and online webstreaming.

Over the years, CityU's Department of Electronic Engineering has been working on this video coding compression technology and its research achievements have been adopted by recognised standard-bearers in the industry. CityU is now at the forefront of the industry and on the verge of creating a new standard to be used around the world.

Courtesy of a government grant, the department recently launched a collaboration with three local universities, Hong Kong Applied Science and Technology Research Institute Company, other government departments and the industry to develop the fifth generation of video coding technology. While there might not be any prizes for guessing the new name - H.265 - CityU researchers will be worthy of kudos if it is endorsed by industry as the new international standard.

Cisco Systems, the world leader in network equipment, projects that 90% of online information will be comprised of video data within three years. To overcome the imminent network overload problems caused by the large volume of video data transmission on the internet, and to enhance video quality, the IT industry has been looking for ways to not only expand network bandwidth but also to improve the compression rate of video files.

Dr Po Lai-man, Associate Professor from the Department of Electronic Engineering at CityU, has been conducting research on the fast motion estimation algorithms central to compression issues for many years and some of his research achievements have been adopted by the International Telecommunication Union. He points out that information transmission has gradually evolved from a text-based mode to a video-based mode over the years.



"Video transmission has become the prevalent form of communication," said Dr Po. "In the old days, Confucius delivered his lectures verbally and his students transcribed his words into written notes and compiled them as the *Analects of Confucius*. Nowadays, many universities videotape their faculty members' lectures and class demonstrations and upload the videos onto websites such as YouTube. This helps disseminate knowledge to a wide audience, near and far."

Dr Po pointed out that the extensive application of video conferencing, video telephoning and digital videotaping already exists. The IT industry's next target will be the improvement of video quality. YouTube, for example, has actually moved towards high definition video. However, the constraints of video quality and file size still pose a challenge in terms of broadcasting, saving and viewing videos - which is where the new standard being devised by CityU researchers come in.

There has been a steady development of video coding technology over the

past 20 years. For example, the current standard H.264 is now being used for high definition broadcasting by local TV stations. The compression of TV signals by H.264 video coding technology makes it possible for TV stations to provide better quality images with existing bandwidth and save space when broadcasting standard definition or high definition TV programmes.

"The new H.265 Standard improves upon the H.264 technologies, upgrading compression capability by 50%. The new standard will make reference to the one set by the European International Telecommunication Union and its development will involve many other research institutes around the world. It is hoped a new standard can eventually be implemented and recognised around the world," said Dr Po.

In regards to the development of the new standard, CityU is responsible for developing the fast motion estimation algorithm. Dr Po explained that the algorithm can save space for video signals by omitting the repeated inputs of identical image data that occurs during transmission.

The collaborative research project will last for 18 months. Dr Po said it was difficult to assess whether a breakthrough in the new technology could be achieved in such short period of time, but the collaboration would definitely enhance technology exchange and improve local research standards. ■



新世代視頻編碼

幻化五彩光影



▶ 人人都說臨考抱佛腳辛苦，不妨試試透過光纖多壓縮50%資訊，你就知道甚麼叫難了。雖說H.264標準之類的專業術語有點讓人摸不着頭腦，卻與現代人的生活息息相關，香港城市大學（城大）的研究人員並將提升這一技術。

H.264是一種快速運動估計算法，並是視像會議、視像電話、網上觀看短片和影片目前所依賴的關鍵視頻編碼壓縮技術。

城大電子工程學系一直致力研究視頻編碼壓縮技術，部分成果已為業界及業內釐定標準的機構所採用，去年更獲政府撥款，首次與本地其他三家大學攜手，與香港應用科技研究院有限公司等政府部門和業界合作，共同研發第五代視頻編碼技術（即

H.265技術標準）。一旦獲得確認，將成為國際認可的新標準。

據全球領先的網絡設備供應商思科系統預測，三年後，互聯網上交流的資訊，將有九成是視訊數據。資訊科技界除了要研究如何擴大頻寬，還必須面對將視訊檔案壓縮得更小的挑戰，以解決因觀看影片而出現網上擁塞的情況，提高視訊質素。

城大電子工程學系副教授布禮文博士多年來一直從事快速運動估計算法研究，有些成果獲得國際電訊聯盟採用。他表示，經過數千年的演變，資訊傳遞方式已從文字逐漸轉變為視訊。

「視訊的傳遞愈來愈重要，」布博士說。「孔子的《論語》由孔子口述、學生筆錄而成；時至今日，外國許多著名大學已將教授講課和課堂上的示範拍成視訊影片，上載到YouTube之類的網站上，任何人都可以瀏覽，使知識傳播得更廣、更遠。」

布博士指出，視像會議、視像電話、數碼攝錄等設備目前已得到廣泛使用，下一步要追求的是質素。視訊設備無論在廣播、存儲及播放上仍須面對畫面質素或檔案容量的限制，甚至YouTube也已開始追求高清質素。



過去20年間，視頻編碼技術規格不斷提升。以高清電視廣播為例，目前使用的最新規格是H.264，本港無線電視的高清台就使用這種規格播放節目。視頻編碼可將電視訊號壓縮，以便電視台使用現有頻寬提供質素較高的圖像，或騰出儲存空間播放較多標清頻道乃至高清頻道。

「H.265標準將保留H.264的某些技術，並改進相關技術，例如，壓縮能力將比H.264高出50%。這個標準的釐定，參考了歐洲國際電訊聯盟的標準；世界各地其他科研機構也會參與研發，希望最終能找到最理想的方案，形成公認標準，」布博士懷着憧憬地說。

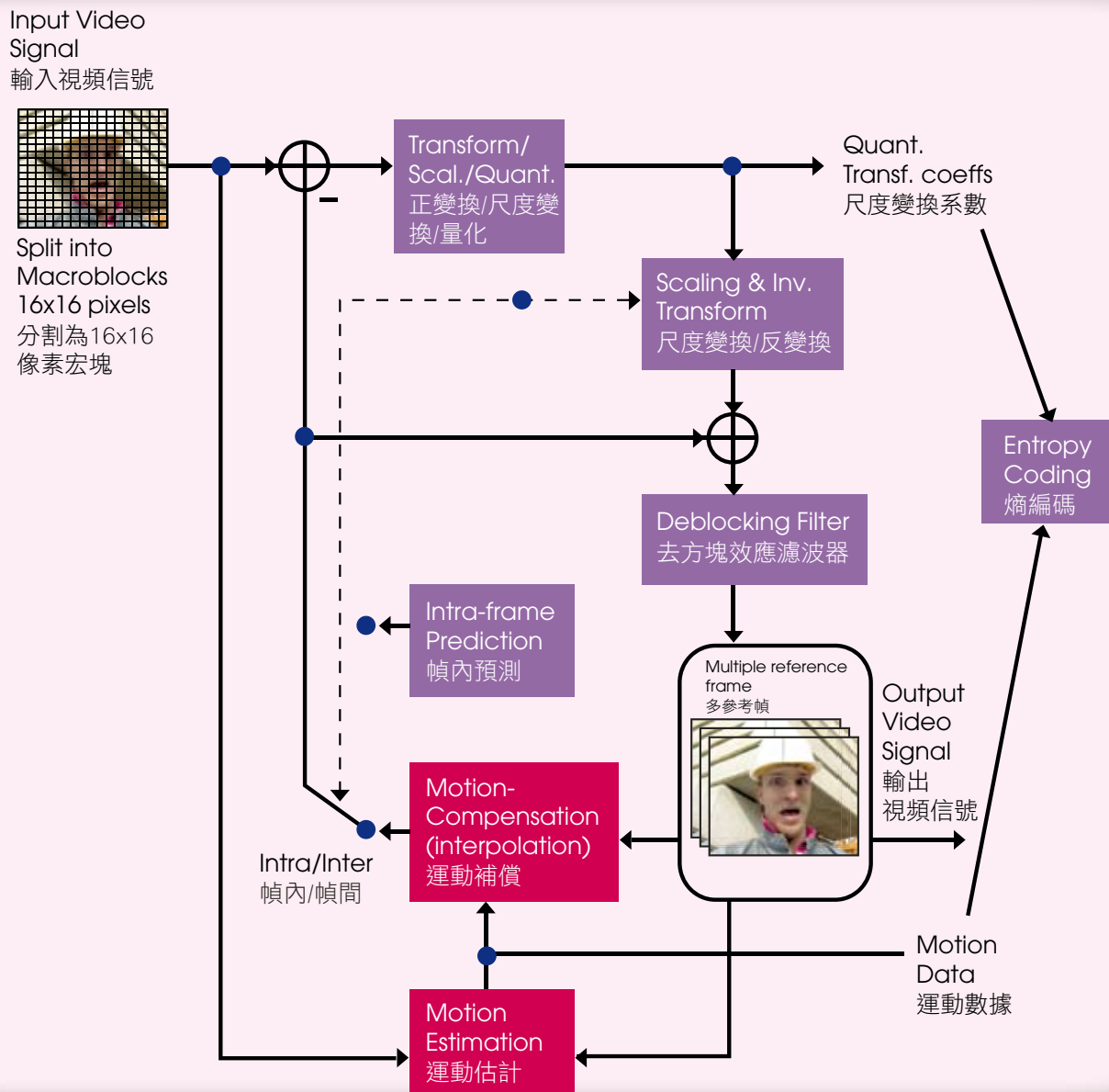
參與研究項目的布博士解釋道，視訊由不同影像組合而成，若前後兩幅有部分相同，可以將前者的相同處複製到後者上，而無須重複輸入數據，以便節省空間，這種技術稱為快速運動估計算法。城大在合作項目中就是負責研究快速運動估計算法。

這次合作項目為期18個月。布博士坦言，能否在短時間內建立突破性新技術，尚無法確定，但合作無疑可促進交流，提高本地研究水平。■

- 1. H.264是視像會議目前所依賴的關鍵視頻編碼壓縮技術。

H.264/AVC Video Encoder Block Diagram

H.264/AVC視頻編碼流程圖



Similar to former video standards, H.264 is also based on hybrid coding framework inside which motion estimation is the most important part in exploiting the high temporal redundancy between successive frames

H.264類似以前的視頻標準，基本以混合編碼框架為依據，其中運動估計又是最重要的部分，因為它充分利用了連續幀之間出現的時間冗餘。